Powder Bed Fusion Development of Oxide Dispersion Strengthened Alloys
Statement of Work (SOW)

March 2026

The National Aeronautics and Space Administration (NASA), through Marshall Space Flight Center (MSFC), requires advanced process development and parameter optimization for laser powder bed fusion (L-PBF) of specialty oxide dispersion strengthened and burn-resistant nickel-based alloys. These include novel NASA-developed alloys such as ORCAlloy-14, ORCAlloy-21, GRX-810, Alloy 754-ODS, OR-F1, and other oxide dispersion strengthened (ODS) compositions. These materials were formulated to enhance high-temperature mechanical performance in extreme environments and/or resist failure modes in oxygen-based combustion and propulsion systems. This current work shall include iterations of the alloy development, including L-PBF parameter optimization with variations in alloy design, preliminary print trials and component builds in partnership with NASA Glenn Research Center (GRC) and MSFC. Additional work includes formal component build development, powder reuse studies, parameter refinement, geometry evaluation, and mechanical validation aligned with NASA technical standards such as NASA-STD-6030.

Task 1, L-PBF Parameter Development and Builds for Burn Resistant Alloys

1. The vendor shall provide access to an EOS M280 laser powder bed fusion system, fully dedicated for developmental use of advanced Ni-based burn-resistant alloys.
2. Target alloys for development include ORCAlloy-14, ORCAlloy-21, Alloy 754-ODS, GRX-810, OR-F1, and additional ODS-based experimental alloys formulated by NASA. Commercially available alloys may be included for benchmarking.
3. The machine shall be available at a minimum operational window of 432 hours/month, with a 60% minimum print utilization rate across the 9-month development period.
4. The vendor shall implement and validate proprietary parameter sets developed in collaboration with NASA MSFC and GRC. These parameters shall yield ≥99.9% relative density and be validated through both microstructural analysis and mechanical testing, subject to NASA approval.
5. The vendor shall develop and optimize new parameter sets for remaining alloys based on NASA-supplied powder specifications and build geometry. Print trials will include test coupons, mechanical specimens, and representative components.
6. The vendor shall document and report any anomalies, build errors, or powder changes during production to NASA in real time.
7. Powder removal, build plate cleaning, and build log tracking (including powder reuse cycles) are the responsibility of the vendor. Powder usage shall be tracked on a best-effort basis to support potential powder aging/recycling studies.
8. Build records, including process parameters, environment controls, and powder certifications, shall be archived and submitted to NASA.
9. All excess powder shall be returned to NASA. ODS powder processing (e.g., mechanical alloying) shall be performed by NASA; the vendor shall coordinate with designated NASA technical POCs for powder handling and delivery.

10. NASA will provide all part models and sample geometries required for print execution.

Task 2, Mechanical Testing of NASA AM Alloys

1. The vendor shall perform a minimum of fifty (100) mechanical tests on specimens fabricated during Task 1.
2. Testing shall include tensile (room temperature and elevated temperature as defined by NASA) and low/high-cycle fatigue per NASA-specified requirements.
3. Test methodologies, sample preparation, and acceptance criteria shall be coordinated with and approved by the NASA technical POC prior to testing.
4. All test data and fracture surfaces (when applicable) shall be documented, and specimens returned to NASA if requested.

Task 3, Powder Development for Developmental Builds

1. The vendor shall supply a minimum of 200 lbs of powder for ORCAlloy-14, ORCAlloy-21, Alloy 754-ODS, and other NASA-specified developmental alloys as required for L-PBF trials under Task 1.
2. Powder shall conform to NASA-provided compositions and particle size distributions, and shall be coordinated with NASA POCs before use.
3. Vendor shall provide full certifications for powder traceability, including chemical composition, particle morphology, and batch tracking, and ensure proper storage and handling during the contract period.

Period of Development

Nine (9) months from authority to proceed (ATP)

Shipping

1. Contractor shall ship all deliverables and process development samples to:

   Central Receiving, Bldg 4631
   Marshall Space Flight
   Center Huntsville, AL 35812

Deliverables

1. Components and development samples and parts built under Task 1.
2. Test Reports from samples in Task 2.
3. Powder certifications from Task 3.